Experiment No.: 9

Date of Performance :
Date of Submission :

Aim: Noise and interference analysis in wireless communication.

Required Equipment / Software :

- Computer/laptop with MATLAB installed.
- Wireless transmitter and receiver or a MATLAB-based simulated wireless communication system.
- Observation sheet for recording SNR, interference condition and BER.

Theory :

Noise is an unwanted random disturbance that affects a communication signal. Thermal noise is commonly modeled as Additive White Gaussian Noise (AWGN). Interference is an unwanted signal from another transmitter or electrical source that overlaps the desired signal.

For a received wireless signal:

$$r(t) = s(t) + n(t) + i(t)$$

where s(t) is the desired signal, n(t) is noise, and i(t) is interference. If no interference is present, i(t)=0.

Signal-to-Noise Ratio (SNR) is the ratio of signal power to noise power:

$$SNR(dB) = 10 \log10(Psignal / Pnoise)$$

When both noise and interference are present, Signal-to-Interference-plus-Noise Ratio (SINR) is used:

$$SINR(dB) = 10 \log10(Psignal / (Pnoise + Pinterference))$$

Bit Error Rate (BER) is the ratio of incorrectly detected bits to the total number of transmitted bits. Increasing noise or interference generally increases BER and reduces link reliability.

Procedure :

A random binary sequence is generated in MATLAB and converted to BPSK symbols. AWGN is added to represent noise. A sinusoidal signal is added to represent narrowband interference. The received signal is detected using a zero-threshold detector, and BER is calculated.

Binary Data → BPSK → AWGN / Interference Channel → Detector → BER Analysis

1. Generate a random binary sequence in MATLAB.
2. Convert the bits to BPSK symbols: $0 \rightarrow -1$ and $1 \rightarrow +1$.
3. Select SNR values and generate AWGN for each SNR.
4. Add AWGN to the transmitted signal.
5. Generate a sinusoidal interfering signal and add it to the noisy signal.
6. Detect the received bits using a zero-threshold detector.
7. Compare detected bits with transmitted bits and calculate BER.
8. Repeat for different SNR values.
9. Plot BER versus SNR for noise-only and noise-plus-interference cases.

Observation Table :

S.No.	SNR (dB)	Condition	Interference	Approx. BER
1	0	AWGN only	Absent	0.07865
2	5	AWGN only	Absent	0.00595
3	10	AWGN only	Absent	0.00078
4	15	AWGN only	Absent	0.00001
5	20	AWGN only	Absent	0.00000
6	10	AWGN + interference	Present	0.00319
7	15	AWGN + interference	Present	0.00003

Program :

```
% Noise and Interference Analysis in Wireless Communication
% BPSK system with AWGN and sinusoidal interference
```

```
clc;
 clear;
 close all;

 rng(1); % Repeatable simulation
 N = 1e5; % Number of transmitted bits SNR_dB =
 0:2:20; % SNR values
 A = 1; % BPSK amplitude

 % Generate random binary data
 data = randi([0 1], 1, N);

 % BPSK modulation: 0 -> -1, 1 -> +1
 tx = A * (2*data - 1);

 % Interference parameters
 f_int = 0.07; % Normalized frequency
 int_amp = 0.55; % Interference amplitude n = 0:N-1;
 interference = int_amp*sin(2*pi*f_int*n);

 ber_noise = zeros(size(SNR_dB));
 ber_interference = zeros(size(SNR_dB));

 for k = 1:length(SNR_dB)

        snr_linear = 10^(SNR_dB(k)/10);
        sigma = sqrt(1/(2*snr_linear));
        noise = sigma*randn(1,N);

        % Noise only
        rx1 = tx + noise;
detected1 = rx1 >= 0;

ber_noise(k) = mean(detected1 ~= data);

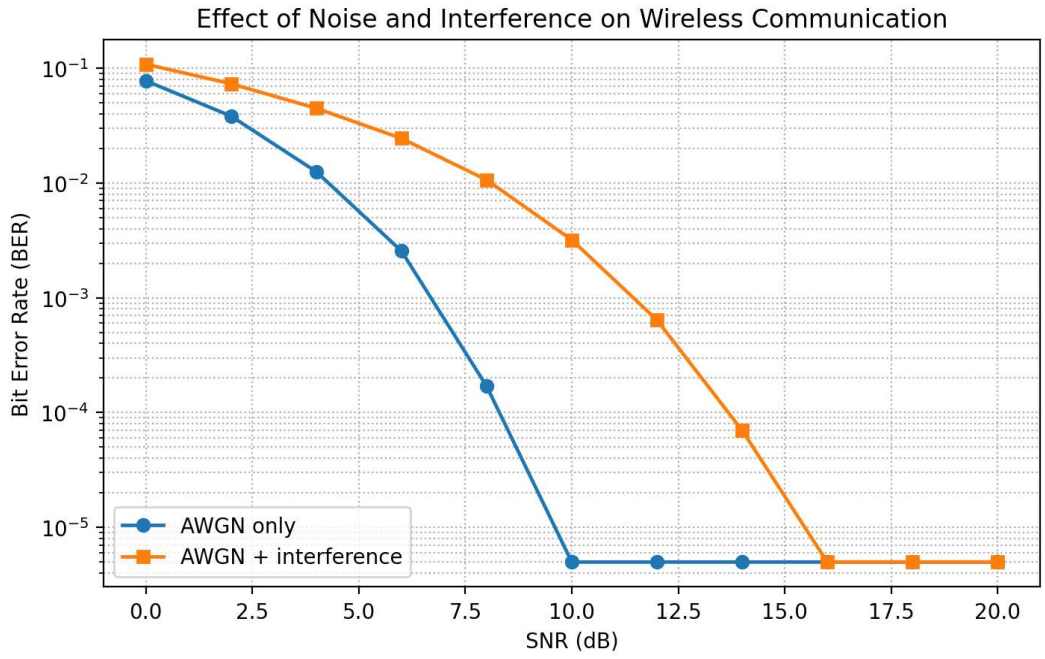
% Noise + interference
        rx2 = tx + noise + interference;
        detected2 = rx2 >= 0;
```

```
        ber_interference(k) = mean(detected2 ~= data);
end

% Display results
T = table(SNR_dB',ber_noise',ber_interference',...
'VariableNames',{'SNR_dB','BER_Noise_Only','BER_Noise_Interference'});
disp(T);

% BER plot
figure;
semilogy(SNR_dB,ber_noise,'-o','LineWidth',1.5);
hold on;
semilogy(SNR_dB,ber_interference,'-s','LineWidth',1.5);
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('Effect of Noise and Interference on Wireless Communication');
legend('AWGN only','AWGN + Interference','Location','southwest');
```

Output :



Result :

The effect of noise and interference on a wireless communication link was analyzed using a BPSK MATLAB simulation. Increasing SNR improved signal detection and reduced BER. Adding an interfering signal increased BER compared with the AWGN-only case.

Conclusion:

Noise and interference degrade wireless communication performance. AWGN produces random disturbances, while interference introduces additional unwanted signal energy. The MATLAB analysis shows that higher SNR generally improves communication quality, whereas interference can maintain a higher BER even when the noise level is relatively low. Practical systems therefore use filtering, channel selection, coding, diversity, power control and interference-management techniques to improve reliability.

Viva Voce Questions :

1. What is the difference between noise and interference?
2. Why does BER decrease as SNR increases?
3. What is co-channel interference?
4. What is adjacent-channel interference?
5. What happens when interference amplitude is increased?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4Marks)	R4 (3 Marks)	Total (15 Marks)	Signature